Mastering ModusToolbox™ in Your Development Workflow

Lab Manual — Hands-On Exercises

Training Session: [Date] | **Presenter:** [Name]
Prerequisites: ModusToolbox™ 3.7 or later installed, internet access for GitHub

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Before You Begin

Prerequisites Checklist

Verify the following before starting the labs:

✅	Requirement	How to Verify
☐	**ModusToolbox 3.7 or later** installed (3.8 recommended if available)	Open Modus-Shell — if it launches, you're good
☐	**Git** available in Modus-Shell	Run `git --version` — Git is included with the ModusToolbox installation
☐	**Internet access** to GitHub	Run `ping github.com` (required for `git clone` and `make getlibs`)
☐	**Eclipse IDE for ModusToolbox** installed	Available in your ModusToolbox installation
☐	(Optional) **KIT_PSC3M5_EVK** eval board + USB cable	Required only for programming exercises

About Modus-Shell

Modus-Shell is a pre-configured terminal that ships with ModusToolbox on Windows operating systems. It sets up all the environment variables — compiler paths, tool locations, and build system paths — so that `make`, `git`, and ModusToolbox CLI tools work out of the box.

How to launch: Windows Start Menu → **Modus-Shell** (or find it in your ModusToolbox installation folder).

Lab 1: Clone, Resolve Dependencies & Build

Segment: 1 — Working Effectively with Version Control & Git
Tools: Modus-Shell (command line only)

> 💡 **Lab 1 uses Modus-Shell exclusively.** All subsequent labs are done in the Eclipse IDE (or VS Code if that is your preference — lab steps document Eclipse).

Objective

Understand the Git + `make getlibs` workflow — clone a real ModusToolbox project from GitHub, resolve its dependencies, and build it. This is the foundational workflow for every ModusToolbox project you'll encounter in the field.

Background

ModusToolbox projects use a lightweight dependency system built on `.mtb` files — small text files that contain a Git URL and version tag pointing to a library. Here's what you need to know:

Concept	What It Means
`.mtb` **file**	A text file containing a Git URL + version tag pointing to a library.
`deps/` **directory**	YOUR declared dependencies — the `.mtb` files you (or Project Creator / Library Manager) put here. **Committed to Git.**
`bsps/` **directory**	YOUR BSP selection — the `.mtb` file for your target board, plus local BSP configuration files. **Committed to Git.**
`libs/` **directory**	Generated by `make getlibs`. Contains the resolved transitive `.mtb` files (dependencies of your dependencies) AND the generated `mtb.mk` makefile that wires everything into the build. **Never committed to Git.**
`mtb_shared/` **directory**	The actual cloned library source code. Lives ONE LEVEL ABOVE your project and is shared across all projects in the workspace. **Never committed to Git.**
`make getlibs`	The dependency resolver. Reads all `.mtb` files in `deps/` and `bsps/`, resolves the full dependency tree, clones source code into `mtb_shared/`, and generates the build wiring in `libs/`.

`deps/` and `bsps/` contain your declared dependencies — the `.mtb` files you control. `make getlibs` resolves the full dependency tree, generates the build wiring in `libs/`, and clones the actual source code into `mtb_shared/`. You commit your declarations; everything else is reconstructed.

Step-by-Step Instructions

Step 1 — Open Modus-Shell

Launch Modus-Shell from your Windows Start Menu. You should see a terminal prompt with the ModusToolbox environment configured.

Step 2 — Create the training workspace

```
mkdir C:/MTB_Training && cd C:/MTB_Training
```

Expected result: You are now in an empty `C:/MTB_Training/` directory.

Step 3 — Clone the training project

```
git clone https://github.com/ClarkJ-Infineon/mtb-training-blinky
```

Expected result: Git clones the repository. You should see output like:

```
Cloning into 'mtb-training-blinky'...
remote: Enumerating objects: ...
Receiving objects: 100% ...
```

> ⚠️ **Notice:** There is no `mtb_shared/` or `libs/` directory yet — only the source code and dependency declarations were committed to Git.

Step 4 — Enter the project directory

```
cd mtb-training-blinky
```

Step 5 — Inspect the declared dependencies

```
ls deps/
```

Expected result: You should see the project's direct dependency `.mtb` files (e.g., `retarget-io.mtb`). Each file is a simple text pointer to a Git repository and version. Note that the BSP is in `bsps/`, not `deps/`.

To see what's inside an `.mtb` file:

```
cat deps/retarget-io.mtb
```

Expected result: A line containing a URL, version tag, and local path — for example:

```
https://github.com/cypresssemiconductorco/retarget-io#release-v1.9.0#$$ASSET_REPO$$/retarget-io/release-v1.9.0
```

Step 6 — Resolve dependencies with `make getlibs`

```
make getlibs
```

Expected result: The tool reads all `.mtb` files in `deps/`, clones each library into `../mtb_shared/`, and generates `libs/mtb.mk`. You'll see Git clone progress for each dependency. This may take 1–3 minutes depending on your internet connection.

When complete, you should see output ending with something like:

```
Library Manager: Completed
```

Step 7 — Inspect the `libs/` directory

```
ls libs/
```

Expected result: You should see many more `.mtb` files than were in `deps/` — these are the **transitive dependencies** (dependencies of your dependencies). You'll also see `mtb.mk` and other generated `.mk` files.

Step 8 — Inspect `libs/mtb.mk`

```
cat libs/mtb.mk
```

Expected result: A series of `SEARCH_*` variable definitions, each pointing to a library path under `../mtb_shared/`. These paths are how the build system finds library source code and headers. For example:

```
SEARCH_retarget-io=../mtb_shared/retarget-io/release-v1.9.0
SEARCH_mtb-hal-psc3=../mtb_shared/mtb-hal-psc3/release-v1.1.2
...
```

Step 9 — Build the project

```
make build
```

Expected result: The GCC ARM compiler builds the project targeting `KIT_PSC3M5_EVK`. Build output scrolls by, and the final lines should indicate success:

```
================================================================================
= Build complete =
================================================================================
```

The build output (`.elf`, `.hex` files) is placed in the `build/` directory.

Results Summary

Step	What You Should See
Clone	Project files appear — `Makefile`, `main.c`, `deps/`, `bsps/`
`ls deps/`	A few `.mtb` files (direct dependencies)
`make getlibs`	Libraries cloned, `libs/mtb.mk` generated
`ls libs/`	Many `.mtb` files (transitive deps) + `mtb.mk`
`cat libs/mtb.mk`	`SEARCH_*` paths pointing to `../mtb_shared/`
`make build`	Successful build with zero errors

Exploration Questions

1. **Transitive dependencies:** Notice that `deps/` has only a few `.mtb` files, but `libs/` has many more — those are transitive dependencies resolved by the build system. What would happen if you deleted `libs/` and `mtb_shared/` and ran `make getlibs` again?
2. **`.mtb` file contents:** Look at the `.mtb` file contents — what three pieces of information does each file contain? (Hint: URL, version, local path.)
3. **Reproducibility:** If a colleague clones this same repo on a different machine and runs `make getlibs` → `make build`, will they get the same result? Why or why not?

Lab 2: Import into Eclipse and Export with mtbarchive

Segment: 2 — Sharing, Importing & Exporting Projects
Tools: ModusToolbox Eclipse IDE

Objective

Import the cloned project into the Eclipse IDE and create a smart archive using `mtbarchive`. This demonstrates the two most common project-sharing workflows: IDE collaboration and archive-based handoff.

Background

IDE Import: ModusToolbox projects are IDE-agnostic — they're built with Makefiles, not IDE project files. To work in Eclipse, you "import in-place," which registers the existing project folder with Eclipse without copying or restructuring anything.

mtbarchive: A ModusToolbox-aware archiving tool that creates self-contained project ZIPs. Unlike a plain ZIP, `mtbarchive` knows which files are authored (include them) and which are generated (exclude them). The result is a small, clean archive that any recipient can restore with `make getlibs` → `make build`.

✅ Included by mtbarchive	❌ Excluded by mtbarchive
Source code, Makefile, configs	`libs/` (regenerated by `make getlibs`)
`deps/` (`.mtb` dependency declarations)	`build/` output
`bsps/` (BSP configuration & device config)	`mtb_shared/` (regenerated)
`.git/` history (if present)	IDE workspace files
README, LICENSE	

Prerequisites

- **Lab 1 completed** — project cloned, dependencies resolved, and built in `C:/MTB_Training/mtb-training-blinky`

Step-by-Step Instructions

Step 1 — Launch ModusToolbox Eclipse IDE

Open **Eclipse IDE for ModusToolbox** from your Start Menu or ModusToolbox installation.

When prompted for a workspace location, enter:

```
C:/MTB_Training
```

Click **Launch**.

Expected result: Eclipse opens with the workspace set to your training directory. The Quick Panel should be visible on the left or bottom of the screen.

Step 2 — Import the project in-place

From the **Quick Panel**, click **Import ModusToolbox™ Application In-Place**.

In the dialog, browse to and select the `mtb-training-blinky` folder.

Expected result: Eclipse registers the project. It appears in the **Project Explorer** panel with source files, BSP, and dependency structure visible.

> 💡 **"In-Place" means Eclipse uses your existing project folder** — it doesn't copy files into the workspace. Your command-line builds and IDE builds operate on the same files.

Step 3 — Configure the ModusToolbox version

Right-click the project in **Project Explorer** → **Properties** → **ModusToolbox** → select/verify the installed ModusToolbox version.

Expected result: The correct ModusToolbox version (e.g., 3.7) is selected and associated with this project.

> ⚠️ **Why is this necessary?** Starting with Eclipse 2026.1, projects must be explicitly associated with an installed ModusToolbox version. If this step is skipped, builds and tool launches may fail or use an unexpected version.

Step 4 — Verify the project in Eclipse

Expand the project in Project Explorer. You should see:

- `main.c` (your application code)
- `deps/` (your `.mtb` dependency declarations)
- `bsps/` (BSP configuration files)
- `libs/` (resolved dependencies — generated)
- `Makefile`

Step 5 — Build from Eclipse

From the **Quick Panel**, click **Build Application** (or right-click the project → Build).

Expected result: The build console shows compilation output, ending with a successful build message — the same result you saw from the command line in Lab 1.

Step 6 — Launch mtbarchive

From the **Quick Panel**, look under **Tools** and click **ModusToolbox archiving and sharing**.

Expected result: The mtbarchive GUI opens with the current project pre-selected.

> 💡 **CLI alternative:** The `mtbarchive` command is not on the PATH by default in Modus-Shell or the IDE terminal. If you prefer the CLI, use the full path:
>
> `C:/Users/<user>/ModusToolbox/tools_3.7/mtbarchive/mtbarchive.exe -v -o project_archive.zip -f .`

Step 7 — Export the archive

In the mtbarchive GUI, use the default settings and click **Export** (or **Create Archive**).

Expected result: An archive ZIP file is created. Note the file size — it should be approximately **1–3 MB**.

Step 8 — Inspect the archive contents

Open the archive with any ZIP viewer (or extract it to a temporary folder). Verify the contents include:

- ✅ Source code (`main.c` , etc.)
- ✅ `deps/` with `.mtb` files
- ✅ `bsps/` with BSP configuration
- ✅ `Makefile`
- ❌ No `libs/` directory
- ❌ No `build/` directory
- ❌ No `mtb_shared/` directory

Results Summary

Step	What You Should See
Eclipse workspace	Set to `C:/MTB_Training`
Import In-Place	Project visible in Project Explorer
Project Properties	Correct ModusToolbox version selected
Build from Eclipse	Successful build (same as command line)
mtbarchive export	ZIP file ~1–3 MB
Archive contents	Source + deps + bsps — no libs, build, or mtb_shared

Exploration Questions

1. **Archive size:** Compare the archive size (~1–3 MB) to the total project size on disk after building (check with `du -sh .` or look at folder properties). Why is the archive so much smaller?
2. **Recipient workflow:** What would a recipient need to do after unzipping this archive to get a working, buildable project? (Hint: two commands.)
3. **Git vs. mtbarchive:** When would you recommend a customer use Git for sharing versus `mtbarchive`? What are the trade-offs?

Lab 3: Create a Custom BSP

Segment: 4 — Workflows to Develop & Utilize a Custom BSP
Tools: Eclipse IDE, Library Manager, Device Configurator

Objective

Create a custom Board Support Package (BSP) from the kit BSP, modify pin assignments, and build with the new BSP. This is the exact workflow you'll guide customers through when they move from an eval board to their production hardware.

Background

Why custom BSPs matter: Infineon kit BSPs are a starting point — they provide a working configuration for the eval board. But most production designs have different pin assignments, crystal frequencies, debug interfaces, and memory layouts. Creating a dedicated custom BSP gives the customer a clean baseline for their specific hardware.

The workflow:

Step	Action
1	Copy the closest kit BSP
2	Rename to `TARGET_<your_name>`

Step	Action
3	Set it as the active BSP (via Library Manager)
4	Edit `design.modus` for your hardware (Device Configurator)
5	Build, test, iterate

Key BSP files:

File	Purpose	Tool
`config/design.modus`	Pin assignments, clock tree, peripheral config	Device Configurator
`config/GeneratedSource/`	Auto-generated C code from `design.modus`	Device Configurator (auto)
`cybsp.h`	Board-level macros (`CYBSP_USER_LED` , `CYBSP_DEBUG_UART` , etc.)	Auto-generated by Device Configurator
`cybsp.c`	Board initialization implementation	Text editor (rarely modified)
`linker/*.ld`	Memory regions	Text editor (advanced)

> ⚠️ **Important:** Do not manually edit files in `GeneratedSource/` — they are regenerated from `design.modus` every time you save in Device Configurator. Your edits would be overwritten.

Prerequisites

- **Lab 1 completed** — project cloned and built in `C:/MTB_Training/mtb-training-blinky`

Step-by-Step Instructions

Step 1 — Copy the kit BSP to create a new BSP

In **Eclipse Project Explorer**, expand `bsps/` in your project. Right-click `TARGET_APP_KIT_PSC3M5_EVK` → **Copy**, then right-click `bsps/` → **Paste**. Rename the pasted folder to `TARGET_MY_BOARD` .

Expected result: A new directory `bsps/TARGET_MY_BOARD/` appears in Project Explorer, containing an exact copy of the kit BSP — `config/design.modus` , `cybsp.h` , `cybsp.c` , `GeneratedSource/` , linker scripts, and all other BSP files.

> 💡 **CLI alternative (IDE integrated terminal):** You can also do this from the terminal within Eclipse or VS Code (see Appendix A for terminal setup):
>
> ```
> cd C:/MTB_Training/mtb-training-blinky
> cp -r bsps/TARGET_APP_KIT_PSC3M5_EVK bsps/TARGET_MY_BOARD
> ```

> 💡 **Naming convention:** Custom BSP directories always start with `TARGET_` . Use a descriptive name — e.g., `TARGET_MY_PRODUCT_V2` , `TARGET_CUSTOMER_MAIN_BOARD` .

Step 2 — Set the new BSP as active using Library Manager

Open **Library Manager** from the Eclipse **Quick Panel** → **Library Manager**.

> 💡 **CLI shortcut (IDE terminal):** You can also launch Library Manager from the IDE's integrated terminal by running `make modlibs` .

In Library Manager:

1. Find `MY_BOARD` in the BSP list
2. Select it as the **active BSP**
3. Click **Update**

Expected result: Library Manager updates the Makefile `TARGET=` variable to `MY_BOARD` , regenerates `libs/mtb.mk` , and resolves dependencies for the new BSP.

Two ways to switch the active BSP:

Option A — Library Manager GUI (recommended)

Select the BSP → click **Update**. Library Manager handles everything in one step.

Option B — Makefile edit + make getlibs (CLI alternative)

1. Open `Makefile` and change `TARGET=APP_KIT_PSC3M5_EVK` to `TARGET=MY_BOARD`
2. Run `make getlibs` to regenerate `libs/mtb.mk` with the new BSP path

Step 3 — Open Device Configurator on the new BSP

From Eclipse, open Device Configurator on the `MY_BOARD` BSP:

- Navigate to `bsps/TARGET_MY_BOARD/config/design.modus`
- Double-click to open in Device Configurator

Alternatively, from the Quick Panel → **Device Configurator** (it should target the active BSP).

Expected result: Device Configurator opens showing the inherited kit configuration — all the pin assignments, clock settings, and peripheral configurations from the original `KIT_PSC3M5_EVK` BSP.

Step 4 — Change one pin assignment

In Device Configurator, modify a pin assignment — for example, move `USER_LED` to a different GPIO pin:

1. Find the current `USER_LED` pin assignment in the Pins tab
2. Disable or reassign it
3. Assign `USER_LED` to a different available GPIO pin

Expected result: The pin assignment updates visually in Device Configurator. The old pin is freed, and the new pin shows the `USER_LED` assignment.

Step 5 — Save the configurator changes

Click **Save** in Device Configurator (or press `Ctrl+S`).

Expected result: Device Configurator regenerates all files in `config/GeneratedSource/`:

- `cycfg_pins.h` / `cycfg_pins.c` — updated pin mux configuration
- `cycfg_clocks.h` / `cycfg_clocks.c` — clock tree configuration
- `cycfg_peripherals.h` / `cycfg_peripherals.c` — peripheral configuration
- `cycfg_system.h` / `cycfg_system.c` — system/power configuration

You can verify the pin change by opening `config/GeneratedSource/cycfg_pins.c` and looking for the new pin assignment.

Step 6 — Build with the custom BSP

```
make build
```

Expected result: The project builds successfully with your custom `MY_BOARD` BSP. The build output should show `TARGET=MY_BOARD` and complete with zero errors.

```
================================================================================
= Build complete =
================================================================================
```

Step 7 — (Optional) Program and verify

If you have a `KIT_PSC3M5_EVK` eval board connected:

```
make program
```

Expected result: The firmware is flashed to the board, and the LED blinks on the new pin (if you wired an LED there, or if the new pin has an on-board LED connected).

Results Summary

Step	What You Should See
Copy BSP	`bsps/TARGET_MY_BOARD/` directory with full BSP contents
Library Manager	`MY_BOARD` set as active BSP, Makefile updated
Device Configurator	Inherited kit configuration, editable
Pin change	New pin assignment visible in configurator
Save	`GeneratedSource/` files regenerated (including `cybsp.h`)

Step	What You Should See
`make build`	Successful build with `TARGET=MY_BOARD`

Exploration Questions

1. **Customer scenario:** A customer sends you their custom board schematic. What's the first thing you do to create their BSP? What questions do you ask the customer before starting?

   Consider:
   - Which eval board did they base their design on?
   - Which silicon variant (MPN) are they using?
   - What pins are reassigned compared to the eval board?
   - What external components are different (crystal, flash, debug interface)?
2. **BSP Assistant vs. manual copy:** When would you use BSP Assistant instead of a manual copy? (Hint: think about changing the target device/MPN.)
3. **Sharing the BSP:** If this custom BSP needs to be shared across multiple projects in a team, what are your options? (Hint: project-local, shared Git repo, or manifest-published.)

Appendix A: ModusToolbox Command Reference

A quick reference for the most commonly used ModusToolbox commands. Run all commands from **Modus-Shell** in your project directory.

Build Commands

Command	What It Does	When to Use
`make build`	Compile the project for the active BSP and toolchain	After code changes, to verify everything compiles
`make clean`	Delete all build artifacts (`build/` directory)	When builds behave unexpectedly, or before a fresh build
`make program`	Build (if needed) and flash firmware to connected board	To deploy and test on hardware
`make getlibs`	Download/update all dependencies from `.mtb` files	After cloning a repo, or when dependencies change

IDE Integration Commands

Command	What It Does	When to Use
`make eclipse`	Generate Eclipse project files	Setting up Eclipse for an existing project
`make vscode`	Generate VS Code settings and `.code-workspace` file	Setting up VS Code without the MTB extension

Tool Launcher Commands

Command	What It Does	When to Use
`make modlibs`	Launch Library Manager	Adding, updating, or removing middleware libraries
`make bsp-assistant`	Launch BSP Assistant	Creating or modifying a BSP (especially MPN changes)

Standalone Tools

Tool	What It Does	When to Use
`mtbarchive-gui`	Launch from Eclipse Quick Panel → **ModusToolbox archiving and sharing** (under Tools)	Sharing a project via email, USB, or support ticket
`mtbarchive` (CLI)	Create a smart project archive from the command line. **Requires full path** — not on the PATH by default	When you prefer the CLI over the GUI

> ⚠️ **CLI path note:** `mtbarchive` is not available directly in Modus-Shell or the IDE terminal. Use the full path:
>
> `C:/Users/<user>/ModusToolbox/tools_3.7/mtbarchive/mtbarchive.exe -v -o <file.zip> -f <project>`

Switching the Active BSP (CLI)

Edit the Makefile `TARGET=` variable, then run `make getlibs` to regenerate `libs/mtb.mk`:

```
# 1. Edit Makefile: change TARGET=APP_KIT_PSC3M5_EVK to TARGET=MY_BOARD
# 2. Regenerate dependency paths
make getlibs
```

Useful Options

Option	Applies To	What It Does
`TOOLCHAIN=GCC_ARM`	`make build`	Specify compiler (default: `GCC_ARM`; also `ARM`, `IAR`)
`CONFIG=Release`	`make build`	Build in Release mode (default: `Debug`)
`TARGET=<BSP_NAME>`	`make build`	Override the active BSP target (use Library Manager instead)

Using Modus-Shell from the IDE

You can run ModusToolbox CLI commands directly from within your IDE without switching to a separate Modus-Shell window.

Eclipse IDE:

1. Go to **Window** → **Show View** → **Terminal** (if not visible, use **Window** → **Show View** → **Other…** → search for "Terminal")
2. The terminal opens with the Modus-Shell environment pre-configured
3. If the terminal does not appear, try **Window** → **Reset Perspective** to restore default views

> 💡 The Eclipse integrated terminal provides the same Modus-Shell environment as the standalone Modus-Shell application — `make`, `git`, and all ModusToolbox CLI tools work out of the box.

VS Code (with ModusToolbox™ extension):

1. Open the integrated terminal (**Terminal** → **New Terminal** or `Ctrl+`` `)
2. Click the **+** (dropdown arrow) next to the terminal tab to see available launch profiles
3. Select **ModusToolbox Shell** from the list
4. A new terminal opens with the ModusToolbox environment configured

> 💡 The ModusToolbox Shell profile is provided by the **ModusToolbox™ Assistant** VS Code extension. If it does not appear, verify the extension is installed and enabled.

Appendix B: Troubleshooting

`make getlibs` Fails

Symptom	Cause	Fix
`fatal: unable to access` or timeout errors	No internet / corporate proxy blocking GitHub	Check network connection; configure Git proxy: `git config --global http.proxy http://proxy:port`
`Permission denied (publickey)`	SSH key not configured for GitHub	Use HTTPS URLs in `.mtb` files (default), or configure SSH keys
`make: *** No rule to make target 'getlibs'`	Not in a ModusToolbox project directory, or Modus-Shell not configured	Verify you're in the project root (where `Makefile` is); restart Modus-Shell

Build Fails After Switching BSPs

Symptom	Cause	Fix
Missing header files (`No such file or directory`)	`libs/mtb.mk` is stale — still pointing to old BSP's paths	Run `make getlibs` to regenerate, or use Library Manager to switch BSPs
`TARGET` mismatch errors	Makefile `TARGET=` doesn't match BSP directory name	Use Library Manager (not manual edit) to switch BSPs
Linker errors referencing wrong memory regions	Stale build artifacts from previous BSP	Run `make clean` then `make build`

Missing Headers After Clone

Symptom	Cause	Fix
`fatal error: cybsp.h: No such file or directory`	`make getlibs` was not run after cloning	Run `make getlibs` then `make build`
`fatal error: cy_pdl_config.h: No such file or directory`	Same — dependencies not resolved	Run `make getlibs` then `make build`
Build works on one machine but not another	Different library versions or missing `make getlibs`	Ensure both machines run `make getlibs`; pin versions in `.mtb` files

Device Configurator Issues

Symptom	Cause	Fix
Changes in Device Configurator don't take effect	Forgot to save, or editing wrong BSP	Verify you're editing the **active** BSP's `design.modus`; save (`Ctrl+S`)
`GeneratedSource/` files look wrong or outdated	Stale generated files	Re-save `design.modus` in Device Configurator to force regeneration
`cybsp.h` macros don't match pin config	`cybsp.h` not updated after Device Configurator change	Manually update `cybsp.h` macros to match new pin assignments

General Tips

- **When in doubt, clean and rebuild:** `make clean && make getlibs && make build`
- **Check `libs/mtb.mk` exists:** If it's missing, `make getlibs` hasn't been run.
- **Verify active BSP:** Open `Makefile` and check the `TARGET=` line — it should match your intended BSP directory name (without the `TARGET_` prefix in the Makefile, but WITH `TARGET_` in the `bsps/` directory name).
- **Read the library README:** If a newly added library doesn't work, check `mtb_shared/<library>/<version>/README.md` for required `COMPONENTS` and `DEFINES`.

Appendix C: Key Concepts Reference

The ModusToolbox Project Structure

```
my-project/
├── Makefile                    ← Project build configuration
├── main.c                      ← Your application code
├── deps/                       ← YOUR declared dependencies (committed to Git)
│   └── retarget-io.mtb
├── bsps/                       ← BSP files (local to project)
│   └── TARGET_APP_KIT_PSC3M5_EVK/
│       ├── cybsp.c / cybsp.h   ← Board init API
│       └── config/
│           └── design.modus    ← Device Configurator master file
├── libs/                       ← Resolved dependency tree (generated, never commit)
│   ├── mtb.mk                  ← Generated SEARCH paths
│   └── (transitive .mtb files)
├── build/                      ← Build output (never commit)
└── mtb_shared/                 ← SHARED library source (one level up, never commit)
    ├── mtb-hal-psc3/
    │   └── release-v1.1.2/
    ├── retarget-io/
    │   └── release-v1.9.0/
    └── ...
```

`deps/` vs `libs/` vs `mtb_shared/` — Know the Difference

	`deps/` + `bsps/`	`libs/`	`mtb_shared/`
What's inside	`.mtb` files you declared (direct dependencies + BSP)	Resolved transitive `.mtb` files + generated `mtb.mk` build wiring	Actual cloned library source code

	`deps/` + `bsps/`	`libs/`	`mtb_shared/`
Who controls it	You (the developer)	`make getlibs` (the tool)	`make getlibs` (the tool)
Commit to Git?	✅ Yes — these are your declarations	❌ No — reconstructed by `make getlibs`	❌ No — reconstructed by `make getlibs`
Scope	Per-project	Per-project	Per-workspace (shared across projects)

What to Commit, What to Ignore

✅ COMMIT	❌ IGNORE
`Makefile`	`build/`
`main.c` and all source files	`mtb_shared/`
`deps/*.mtb`	`libs/`
`bsps/` (BSP customizations)	`.vscode/`
`.gitignore`	`.cache/`

End of Lab Manual